A Memory-Efficient FPGA Implementation and Hardware/Software Co-Design of Post-Quantum HQC-128 for Resource-Constrained Edge Devices

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***Abstract*—Deploying the National Institute of Standards and Technology (NIST) Hamming Quasi-Cyclic (HQC-128) cryptosystem on edge nodes requires hardware accelerators that fit within tight on-chip memory limits. Most prior full-KEM and decapsulation field-programmable gate array (FPGA) designs consume 17.5 to 62.5 Block RAM (BRAM) tiles, which occupies up to 125% of the 50-BRAM capacity on an AMD Artix-7 xc7a35t. This letter presents a memory-efficient hardware/software co-design for HQC-128 public-key encryption (PKE). An interval conflict graph of polynomial lifetimes ($\chi(G) = 6$) fits all 14 variables into 6.0 RAMB36 tiles and 408 distributed lookup table (LUT) RAMs. Three 64-bit ring multipliers time-share a dual-port accumulator alongside a Chien-search and Forney Reed-Solomon codec. On an Artix-7 xc7a35t-1, the routed core occupies 8,246 LUTs, 7,868 flip-flops, 6.0 BRAMs (12.0%), and 0 digital signal processing (DSP) slices. At 119.0 MHz, the core consumes 0.158 W and 118.5 μJ per KeyGen+Encrypt+Decrypt sequence (0.750 ms, 89,262 cycles). Compute cycles drop by $2,949\times$ for PKE ($1,951\times$ for Fujisaki–Okamoto key encapsulation [FO-KEM]) relative to 32-bit RISC-V software.**

***Index Terms*—Edge devices, field-programmable gate array (FPGA), Hamming Quasi-Cyclic (HQC), hardware/software co-design, post-quantum cryptography (PQC).**

I. Introduction

LARGE-SCALE quantum computers running Shor's algorithm solve integer factorization and discrete logarithms in polynomial time and break RSA and elliptic-curve schemes. Alongside the lattice-based ML-KEM standard (NIST FIPS 203), the National Institute of Standards and Technology (NIST) selected Hamming Quasi-Cyclic (HQC) [1], [2] in March 2025 for standardization as a code-based alternative based on syndrome decoding over quasi-cyclic codes. Unlike Classic McEliece (public keys > 260 KB), HQC-128 uses a 2,249-byte public key and bounds the decryption failure rate (DFR) below 2^{-128} for bandwidth-limited edge telemetry. However, HQC-128 operates on binary polynomials of degree $N = 17,669$ in cyclic ring $\mathscr{R} = \mathbb{F}_2[X]/(X^N - 1)$ and decodes concatenated shortened Reed-Solomon (RS) and duplicated Reed-Muller (RM) codes.

On a 32-bit embedded RISC-V processor (CV32E40P), pure software execution of HQC-128 takes 263.3 million cycles (43.6M key generation, 87.6M encapsulation, 132.1M decapsulation) [3], where 17,669-bit polynomial multiplication accounts for ≈95% of runtime. Offloading HQC to an entry-level FPGA System-on-Chip (SoC) such as the AMD Artix-7 xc7a35t (UG473, 20,800 LUTs, 50 BRAM tiles) removes this bottleneck only if the accelerator leaves the vast majority of on-chip BRAM available for host RISC-V memories and queues. Most HQC-128 FPGA cores exceed this budget: Deshpande et al. [4] use 20.0–24.0 BRAMs; Antognazza et al. [5] use 20.0–20.5 BRAMs per primitive; Tu and Xie (LEAH) [6] use 7.0, 12.5, and 17.5 BRAMs across cores; and Wang et al. [7] use 62.5 BRAMs (125% of an xc7a35t). Earlier co-designs such as Schöffel et al. [8] use 3.0 BRAMs and 0 DSPs, but take 1.30 ms for encapsulation and 5.60 ms for decapsulation.

Because edge SoCs include a host CPU for networking, dedicated fabric Keccak sponges are unnecessary. We partition HQC-128 into a HW/SW co-design: the host CPU runs SHAKE256 (FIPS 202) seed expansion and index sampling in software, while a unified FPGA coprocessor executes all three PKE primitives (*KeyGen*, *Encrypt*, *Decrypt*) in 6.0 RAMB36 tiles. Our contributions are:

1) *Memory-Lifetime Conflict Bound:* We model all 14 logical buffers (7 dense polynomials, 3 accumulators, and 4 sparse arrays) as an interval conflict graph G with $\chi(G) = \omega(G) = 6$. In-place aliasing ($\mathbf{u} \leftrightarrow \mathbf{n}$, $\mathbf{v} \leftrightarrow \mathbf{mG}$), a shared accumulator ($\mathbf{T}_1$), and distributed LUTRAM (408 LUTs) reduce storage to 6.0 RAMB36 and 0 RAMB18 tiles.
2) *64-Bit vs. 72-Bit Word-Width Analysis and Codec:* We show analytically why a 64-bit datapath avoids the addressing and alignment penalties of native 72-bit RAMB36 ports for $N = 17,669$ ring folding and $17,664$-bit RS-RM codewords, and integrate 64-bit circular shifters with a Chien-search and pipelined Forney RS decoder verified across 100 NIST vectors.
3) *SAIF-Annotated Power and Energy Scaling:* Post-route power analysis with 100-vector simulation switching activity on Artix-7 xc7a200t-2 and xc7a35t-1 shows that running the xc7a35t-1 core at 119.0 MHz amortizes static power and cuts total PKE energy by 66.5% (to 118.5 μJ) relative to 25 MHz.
4) *AXI4-Lite SoC Co-Design:* An AXI4-Lite bridge (153 LUTs, 293 FFs) and C99 driver interface the core to complete PKE computation in 0.750 ms (89,262 core cycles, or 134,939 with FO re-encryption), leaving 88.0% of xc7a35t BRAMs free.

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TABLE I
POST-ROUTE IMPLEMENTATION AND SAIF POWER METRICS ACROSS ARTIX-7 TARGETS

Metric	Artix-7 xc7a200t-2	Artix-7 xc7a35t-1
Slice LUTs (Logic / Memory)	8,246 (7,838 / 408)	8,246 (7,838 / 408)
Slice Registers (FF)	7,868 (2.9%)	7,868 (18.9%)
Block RAM (RAMB36E1)	**6.0** (1.6%)	**6.0** (**12.0%**)
DSP48E1 Slices	**0** (0.0%)	**0** (0.0%)
Routed WNS (@100.0 MHz)	+2.410 ns ($T_{\text{crit}} = 7.590$ ns)	+1.641 ns ($T_{\text{crit}} = 8.359$ ns)
Post-Route Extrapolated $F_{\max}$ (Sweep)*	131.8 MHz (eval. @ 131.0 MHz)	119.6 MHz (eval. @ 119.0 MHz)
Power @100 MHz (Dyn/Stat)	0.202 W (0.080/0.122 W)	0.146 W (0.078/0.069 W)
Power @Swept $F_{\max}$ (Dyn/Stat)	0.222 W (0.100/0.122 W)	0.158 W (0.089/0.069 W)
PKE Energy (@100 MHz)	180.3 μJ (0.893 ms)	130.3 μJ (0.893 ms)
PKE Energy (@Swept $F_{\max}$)	151.3 μJ (0.681 ms)	**118.5** μ**J** (0.750 ms)

*Extrapolated from routed critical path $T_{\text{crit}} = 10.0\text{ns} - \text{WNS}$; power and latency are evaluated at 131.0 and 119.0 MHz.

II. HQC-128 PRELIMINARIES AND BOTTLENECKS

A. HQC-128 Public-Key Encryption

Let $\mathbb{F}_2$ denote the binary field and $\mathscr{R} = \mathbb{F}_2[X]/(X^N - 1)$ the cyclic ring with prime $N = 17,669$. Bold lowercase letters denote N-bit polynomials in $\mathscr{R}$, and $\mathscr{R}_\omega \subset \mathscr{R}$ denotes sparse polynomials of Hamming weight ω. In HQC-128 [1], secret polynomials use $\omega = 66$ and ephemeral encryption polynomials use $\omega_r = \omega_e = 75$ (DFR $< 2^{-128}$).

PKE.KeyGen: From public $\mathbf{h} \in \mathscr{R}$ and secret sparse $\mathbf{x}, \mathbf{y} \in \mathscr{R}_{66}$, key generation computes $\mathbf{s} = \mathbf{x} + \mathbf{h} \cdot \mathbf{y} \in \mathscr{R}$.

PKE.Encrypt: A 128-bit message $\mathbf{m} \in \mathbb{F}_2^{128}$ is encoded by an outer shortened RS$[n_1 = 46, k_1 = 16, d_1 = 31]$ code over $\mathbb{F}_{2^8}$ and an inner duplicated RM$[n_2 = 384, k_2 = 8, d_2 = 192]$ code (multiplicity 3) to form $\mathbf{mG} \in \mathbb{F}_2^{17,664}$. With ephemeral $\mathbf{r}_1, \mathbf{r}_2, \mathbf{e} \in \mathscr{R}_{75}$, ciphertext $(\mathbf{u}, \mathbf{v})$ is evaluated in $\mathscr{R}$ as $\mathbf{u} = \mathbf{r}_1 + \mathbf{h} \cdot \mathbf{r}_2$ and $\mathbf{v} = \mathbf{mG} + \mathbf{s} \cdot \mathbf{r}_2 + \mathbf{e}$.

PKE.Decrypt: Using secret key $\mathbf{y} \in \mathscr{R}_{66}$, decryption computes $\mathbf{n} = \mathbf{v} - \mathbf{u} \cdot \mathbf{y} = \mathbf{mG} + (\mathbf{x} \cdot \mathbf{r}_2 - \mathbf{y} \cdot \mathbf{r}_1 + \mathbf{e}) \in \mathscr{R}$, and decodes its first 17,664 bits via a Fast Walsh-Hadamard Transform (FWHT) RM decoder and Berlekamp-Massey/Forney RS decoder to recover $\mathbf{m}' \in \mathbb{F}_2^{128}$. In full HQC-128 key encapsulation (KEM), FO decapsulation follows *PKE.Decrypt* with re-encryption (*PKE.Encrypt*).

B. Causes of High BRAM Usage in Existing Hardware

Although one polynomial in $\mathscr{R}$ holds 17,669 bits (2,209 bytes, $< 50\%$ of a 36-Kb RAMB36E1), prior HQC-128 cores consume 17.5–62.5 BRAMs due to three factors: 1) *Wide-Port Fragmentation:* Designs using 128-bit TDP memories [4], [5] stripe each polynomial across four RAMB36E1 tiles ($4 \times 36 \geq 128$), consuming 20.0 BRAMs for five buffers while leaving $> 75\%$ of bit cells unused. 2) *Private Accumulators and Buffers:* Unified designs [4], [7] allocate separate scratchpad BRAMs per primitive and maintain distinct memories for $\mathbf{mG}, \mathbf{v}, \mathbf{u}$, and $\mathbf{n}$. 3) *Shallow Tables in BRAM:* Sparse vectors ($\leq 75 \times 15$-bit $= 1,125$ bits) and $\mathbb{F}_{2^8}$ lookup tables (128–256 bytes) mapped to Block RAM [4], [7] waste over a dozen BRAMs on sub-kilobyte arrays.

III. PROPOSED HW/SW CO-DESIGN ARCHITECTURE

A. Memory-Lifetime Conflict Graph and 6-BRAM Allocation

Across the five PKE phases ($\Phi_{\text{kg}} \to \Phi_{\text{enc-codec}} \to \Phi_{\text{enc-arith}} \to \Phi_{\text{dec-arith}} \to \Phi_{\text{dec-codec}}$), the core manages 14 buffers: 7 dense polynomials $(\mathbf{h}, \mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{v}, \mathbf{mG}, \mathbf{n})$, 3 accumulators $(\mathbf{T}_{\text{kg}}, \mathbf{T}_{\text{enc}}, \mathbf{T}_{\text{dec}})$, and 4 sparse arrays $(\mathbf{y}, \mathbf{r}_1, \mathbf{r}_2, \mathbf{e})$, as shown in Fig. 1.

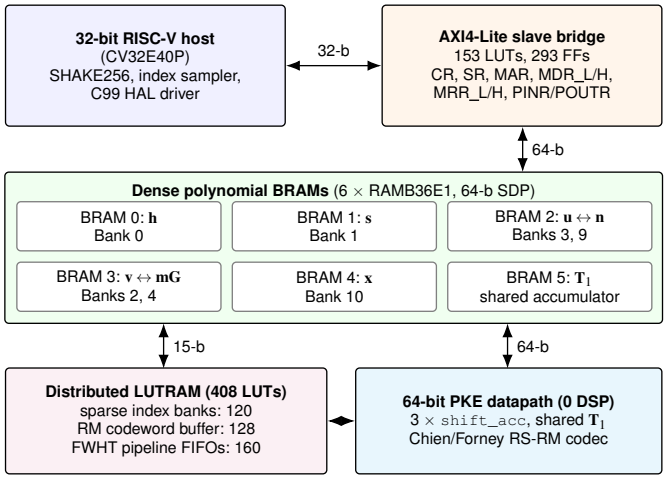


Fig. 1. HQC-128 HW/SW co-design architecture: 32-bit AXI4-Lite bridge, six lifetime-aliased RAMB36E1 polynomial memories, 408-LUT distributed LUTRAM, and the 64-bit PKE datapath.

First, four sparse arrays $\mathbf{y}$ (66×15-bit) and $\mathbf{r}_1, \mathbf{r}_2, \mathbf{e}$ (75×15-bit each) map to distributed LUTRAM, using 120 LUTRAMs (30 per bank). Combined with 128 LUTRAMs for the RM encoder and 160 LUTRAMs for the 7-layer Fast Walsh-Hadamard Transform (FWHT) decoder FIFOs, total distributed memory is 408 LUTRAMs and 0 BRAMs. Dense secret $\mathbf{x}$ resides in BRAM 4; after 64-bit multiplication $\mathbf{h} \cdot \mathbf{y}$ accumulates into $\mathbf{T}_1$ ($66 \times 282 = 18,612$ cycles), *KeyGen* finishes $\mathbf{s} = \mathbf{x} \oplus \mathbf{T}_1$ in 831 cycles (6.98 μs at 119 MHz, 19,796 total cycles) via a 98-LUT word-wise XOR unit.

Second, we model buffer residency across PKE phases as an interval conflict graph $G = (V, E)$. Disjoint arithmetic phases collapse the three ring accumulators into a shared dual-port BRAM $\mathbf{T}_1$, which forms the 8-vertex dense buffer set $V = \{\mathbf{h}, \mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{v}, \mathbf{mG}, \mathbf{n}, \mathbf{T}_1\}$ (277×64-bit each). An edge $(u, w) \in E$ exists iff lifetimes overlap: $I(u) \cap I(w) \neq \emptyset$.

Theorem 1 (Chromatic Memory Lower Bound): *Under persistent on-chip keys $\{\mathbf{h}, \mathbf{s}, \mathbf{x}\}$ with SDP Block RAMs, the chromatic number of conflict graph G satisfies $\chi(G) = 6$. Exactly 6.0 RAMB36E1 tiles is minimal and sufficient for stall-free execution.*

Proof: An interval graph is chordal and therefore perfect. By the Hajnal-Surányi theorem on graph coloring, the chromatic number equals the clique number: $\chi(G) = \omega(G)$. During $\Phi_{\text{enc-arith}}$, keys $\{\mathbf{h}, \mathbf{s}, \mathbf{x}\}$ are resident, ciphertext $\mathbf{u} = \mathbf{r}_1 + \mathbf{h} \cdot \mathbf{r}_2$ is held, and $\mathbf{T}_1$ computes $\mathbf{s} \cdot \mathbf{r}_2$. At step t^*, codeword $\mathbf{mG}$ is read to form $\mathbf{T}_1 \leftarrow \mathbf{T}_1 \oplus \mathbf{mG}$. Here, all six buffers are simultaneously live: $t^* \in I(\mathbf{h}) \cap I(\mathbf{s}) \cap I(\mathbf{x}) \cap I(\mathbf{u}) \cap I(\mathbf{mG}) \cap I(\mathbf{T}_1)$, which forms a 6-clique $K_6 \subseteq G$. Hence $\omega(G) \geq 6$ and $\chi(G) \geq 6$. After t^*, $\mathbf{mG}$ expires ($t_{\text{end}}(\mathbf{mG}) = t^*$). Ciphertext $\mathbf{v} = \mathbf{T}_1 \oplus \mathbf{e}$ commits after t^*, so $I(\mathbf{mG}) \cap I(\mathbf{v}) = \emptyset$; this allows $\mathbf{mG}$ and $\mathbf{v}$ to share BRAM 3. During $\Phi_{\text{dec-arith}}$, $\mathbf{u}$ expires before decoded $\mathbf{n} = \mathbf{v} \oplus \mathbf{T}_1$ overwrites BRAM 2 ($I(\mathbf{u}) \cap I(\mathbf{n}) = \emptyset$). Greedy coloring assigns $\mathscr{C} = \{\mathbf{h} \mapsto 0, \mathbf{s} \mapsto 1, \{\mathbf{u}, \mathbf{n}\} \mapsto 2, \{\mathbf{mG}, \mathbf{v}\} \mapsto 3, \mathbf{x} \mapsto 4, \mathbf{T}_1 \mapsto 5\}$, which confirms $\chi(G) = 6.0$. ■

TABLE II
COMPARISON WITH PRIOR HARDWARE IMPLEMENTATIONS OF HQC-128 ON ARTIX-7 FPGAS

Design	Platform (Device / Speed)	Scope (SHAKE256)	LUTs	FFs	BRAM (36k)	DSP	F_{max} (MHz)	Latency (ms) KG / Enc / Dec (FO-Dec)	Energy
Schöffel et al. [8] (Co-Design)	xc7a100t [NR]	KEM Co-Des. (SW)	∼8,000	2,400	3.0	0	100	0.56 / 1.30 / (5.60)	NR
Deshpande et al. [4] (Balanced)	xc7a100t-3	Full KEM (HW)	13,865	6,897	22.0	8	164	0.10 / 0.20 / (0.29)	NR
Deshpande et al. [4] (HighSpeed)	xc7a100t-3	Full KEM (HW)	15,214	7,293	24.0	8	178	0.09 / 0.13 / (0.21)	NR
Deshpande et al. [4] (Decap Only)	xc7a200t-3	Decap Only (Excluded)	8,984	6,596	20.0	4	192	– / – / (0.251)	NR
Antognazza et al. [5] (KeyGen)	xc7a200t-3*	KeyGen Only (Excluded)	6,827	5,444	20.0	0	208	0.030 / – / –	NR
Antognazza et al. [5] (Encap)	xc7a200t-3*	Encap Only (Excluded)	7,064	6,378	20.0	0	168	– / 0.079 / –	NR
Antognazza et al. [5] (Decap)	xc7a200t-3*	Decap Only (Excluded)	13,327	9,488	20.5	0	167	– / – / (0.119)	NR
Tu & Xie (LEAH) [6] (KeyGen)	xc7a200t [NR]	KeyGen Only (HW)	2,834	1,626	7.0	0	218	0.052 / – / –	NR
Tu & Xie (LEAH) [6] (Encap)	xc7a200t [NR]	Encap Only (HW)	5,455	3,280	12.5	0	199	– / 0.071 / –	NR
Tu & Xie (LEAH) [6] (Decap)	xc7a200t [NR]	Decap Only (HW)	9,878	10,212	17.5	0	191	– / – / (0.129)	NR
Wang et al. [7]	xc7a200t-3	Full KEM (HW)	20,201	13,685	62.5	NR	157	0.079 / 0.114 / (0.207)	NR
This Work (xc7a200t-2 @131 MHz)	**xc7a200t-2**	**Unified PKE (SW)**	**8,246**	**7,868**	**6.0**	**0**	**131.8**	**0.151 / 0.349 / 0.182 (0.530)**	**151.3 µJ**
This Work (xc7a35t-1 @119 MHz)	**xc7a35t-1**	**Unified PKE (SW)**	**8,246**	**7,868**	**6.0**	**0**	**119.6**	**0.166 / 0.384 / 0.200 (0.584)**	**118.5 µJ**

NR = Not Reported. HW/SW indicates on-chip vs. host-CPU SHAKE256; Excluded indicates benchmark without SHAKE256. Dec (FO-Dec) lists PKE.Decrypt and FO-KEM Decaps (Dec + Re-Enc). *Validated on xc7a100t-1.

Boundary Conditions: Theorem 1 holds under persistent keys and stall-free SDP scheduling in a 64-bit layout. The bound shifts under three conditions: 1) *Key Eviction:* If secret $\mathbf{x}$ is evicted to host RAM across runs and reloaded over AXI only for decryption, $\mathbf{x} \notin V$ during encryption. The clique at t^* drops to $\{\mathbf{h}, \mathbf{s}, \mathbf{u}, \mathbf{mG}, \mathbf{T}_1\}$, so $\omega(G) = \chi(G) = 5$ (5.0 BRAMs), but this requires 277 64-bit reloads (831 AXI transactions) and exposes secret keys on the bus. 2) *LUTRAM Secret:* Storing $\mathbf{x}$ in distributed LUTRAM drops BRAMs to 5.0 at the cost of 277 RAM64X1D primitives (554 slice LUTs). 3) *Dual Packing:* In 64-bit mode, two 277-word polynomials require 554 rows in one tile, which exceeds the 512-row depth limit. While a 72-bit tile fits two polynomials ($2 \times 246 = 492 \leq 512$ rows) to reach 5 BRAMs with idle secret $\mathbf{x}$, it incurs the codec misalignment analyzed in Section III-B.

B. 64-Bit vs. 72-Bit Datapath Word-Width Analysis

In SDP mode, a RAMB36E1 supports a 72-bit port (512×72 bits), which takes $\lceil 17,669/72 \rceil = 246$ words per polynomial versus $\lceil 17,669/64 \rceil = 277$ words for 64 bits. However, a 72-bit datapath incurs two functional penalties: *1) Addressing and Cycle Trade-Off:* Decomposing a 15-bit sparse index $\text{pos}[j]$ into word address and bit shift is a 0-LUT wire slice for $W_d = 64$ ($[14:6]$ and $[5:0]$), whereas $W_d = 72$ requires division by 72 (estimated ≈ 45 LUTs). While 72-bit datapaths save 11% cycles (251 vs. 282/multiply), RS-RM codec misalignment negates this gain. *2) RS-RM Codeword Misalignment:* The 17,664-bit RS-RM codeword (46×384 bits) occupies an exact integer $17,664/64 = 276.0$ words (6 words per RM block), whereas $17,664/72 = 245.333$ words splits 384-bit RM blocks across uneven 24/48-bit boundaries. In the 64-bit layout, word 276 holds $17,669 - 276 \times 64 = 5$ residual bits (masked with `0x1F`), which fold directly into word 0 during cyclic shift accumulation.

C. Cyclic Ring Multiplier and Concatenated RS-RM Codec

Polynomial multiplication decomposes into sparse cyclic shifts: $\mathbf{c}(X) = \sum_{j=0}^{W-1} \mathbf{a}(X) \cdot X^{\text{pos}[j]} \bmod (X^N - 1)$ for $W \in \{66, 75\}$. To avoid wide multiplexing, each primitive instantiates a local 64-bit `shift_acc` engine (636, 639, 618 LUTs; 306 FFs each) that time-shares accumulator $\mathbf{T}_1$. For each index $\text{pos}[j]$, the engine latches word offset $\text{pos}[j][14:6]$ and bit shift $s = \text{pos}[j][5:0]$, and streams $\mathbf{a}[0 \ldots 276]$ through a 128-bit funnel shifter ($\{\text{word}, \text{carry}\} \ll s$). Because word 276 holds only 5 valid bits, the datapath stitches $(\text{shifted} \ll 59) \mid (\text{prev_shifted} \gg 5)$ upon address wrap-around, and completes each multiplication in $W \times 282$ cycles ($277 + 5$ pipeline latency cycles) with 0 DSPs.

During *Dec-Codec*, the RM decoder reads three 128-bit words (384 bits) per block. Because BRAM 2 has a 64-bit port, the adapter prefetches six 64-bit words into a 3×128-bit buffer over 8 cycles during the 90-cycle FWHT window, to hide width-conversion latency. In the RS decoder, after 30-syndrome evaluation and Berlekamp-Massey updates, an iterative Chien search over $\mathbb{F}_{2^8}$ and pipelined Forney unit evaluate error roots and magnitudes. Registering Forney $\mathbb{F}_{2^8}$ multiplier inputs across issue and commit states shortens the xc7a35t-1 critical path from 9.844 ns to 8.359 ns (F_{max} from 101.6 to 119.6 MHz) for 15 LUTs, 26 FFs, and 106 cycles. *KeyGen* (19,796 cycles) and *Encrypt* (45,677 cycles) have fixed latency, while RS-RM decoding takes 3,863–4,413 cycles (23,789 average *Decrypt* cycles across 100 vectors).

D. AXI4-Lite Slave Bridge and HAL Driver

The AXI4-Lite bridge (153 LUTs, 293 FFs, 0 BRAMs) interfaces the core with a 32-bit host bus via control (`CR`), status (`SR`), address (`MAR`, packing 4-bit bank select in $[3:0]$ and 9-bit word offset in $[12:4]$), data (`MDR_L/H`, `MRR_L/H`), and plaintext registers (`PINR`/`POUTR`). Writing `MDR_H` pulses a strobe that commits {`MDR_H, MDR_L`} atomically into the targeted bank.

IV. EXPERIMENTAL RESULTS AND COMPARISON

A. Implementation and SAIF-Annotated Power Analysis

The unified HQC-128 PKE core was implemented in Verilog HDL and routed in AMD Vivado 2025.2 targeting the Artix-7 xc7a200t-2 and edge xc7a35t-1 (20,800 LUTs, 50 RAMB36E1 tiles) under a 10.0 ns (100 MHz) clock constraint. Functional verification across 100 NIST Round 4 PQClean test vectors passed with zero mismatches. Post-route power was estimated in Vivado by annotating routed netlists with switching activity from an RTL behavioral simulation across 100 NIST Round 4 test vectors (`hqc_pke_100vec.saif`). The behavioral SAIF matches 5,115 of 19,285 routed nets (27%, spanning primary registers, memory interfaces, and state logic); Vivado vectorless estimation covers unmatched internal combinational nets.

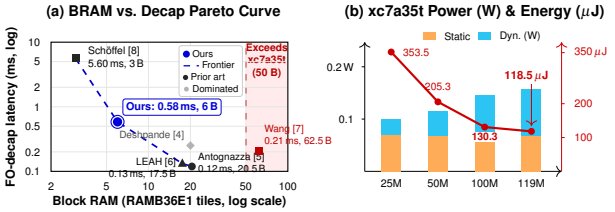


Fig. 2. (a) BRAM vs. FO-decap latency Pareto curve on Artix-7 (shaded area exceeds 50-BRAM xc7a35t limit). (b) Post-route power bars (left) and total PKE energy curve (right) vs. frequency on xc7a35t-1.

Table I and Fig. 2(b) summarize routed metrics across both dies. At 100 MHz, xc7a200t-2 achieves +2.410 ns WNS ($F_{\max} = 131.8$ MHz, swept at 131.0 MHz) and xc7a35t-1 achieves +1.641 ns WNS ($F_{\max} = 119.6$ MHz, swept at 119.0 MHz). Transitioning from xc7a200t-2 to xc7a35t-1 reduces static power by 43.4% (0.122 to 0.069 W), with dynamic power at 100 MHz of 0.078 W (0.146 W total, 130.3 μJ). Across 25–119.0 MHz on xc7a35t-1, total power scales from 0.099 to 0.158 W, while energy per PKE cycle (89,262 cycles) drops by 66.5% (353.5 to 118.5 μJ) because the 0.069 W static leakage integrates over a 4.76× shorter window (0.750 ms vs. 3.570 ms).

B. Comparison with Prior Hardware Implementations

Memory and Logic: As plotted in Fig. 2(a) and Table II, our core defines the lowest-memory sub-millisecond unified PKE design on edge FPGAs: Schöffel et al. [8] use 3.0 BRAMs but take 5.60 ms for decapsulation, whereas sub-millisecond decapsulation and full-KEM cores [4], [5], [6] consume 2.9×–4.0× more BRAMs (17.5–24.0) and Wang et al. [7] (62.5 BRAMs) exceed the 50-BRAM xc7a35t budget entirely. In logic, our core requires 38.1% fewer LUTs than Antognazza et al.'s decapsulation core [5] (13,327 LUTs) and 16.5% fewer LUTs than LEAH's decapsulation core [6] (9,878 LUTs), with 0 DSPs. Compared to Schöffel et al. [8] (3.0 BRAMs, 0 DSPs on xc7a100t), our coprocessor accelerates PKE encryption to 0.384 ms (3.4× faster than their 1.30 ms KEM encapsulation) and FO-decapsulation compute to 0.584 ms (9.6× faster than their 5.60 ms KEM decapsulation). *Throughput Trade-Offs:* Standalone and full-KEM cores achieve higher $F_{\max}$ (157–218 MHz, aided by −3 speed grades in [4], [5], [7]) and lower latencies (0.030–0.100 ms KeyGen, 0.071–0.200 ms Encap, 0.119–0.290 ms FO-Decap) using 128-bit multi-port memories, parallel multipliers, and on-chip SHAKE256, at the cost of 17.5–62.5 BRAMs. Serializing $\mathbf{h}\cdot\mathbf{r}_2$ and $\mathbf{s}\cdot\mathbf{r}_2$ through one 64-bit multiplier and shared $\mathbf{T}_1$ (45,677 cycles) restricts memory to 12.0% of xc7a35t BRAMs.

C. Embedded HW/SW Co-Design Evaluation

Table III compares our coprocessor (at 119.0 MHz on xc7a35t-1) against the official HQC-128 reference software on a 32-bit RISC-V processor (CV32E40P, `-O3`) [3]. In PKE mode, the core completes all three primitives in 89,262 cycles (0.750 ms, 2,949× fewer compute cycles than the 263.3M-cycle baseline). For FO-KEM decapsulation where the core executes *PKE.Decrypt* followed by re-encryption (*PKE.Encrypt*) in the same 6.0 BRAMs, decapsulation takes $23,789 + 45,677 = 69,466$ core cycles (0.584 ms, 1,902× reduction) and total KEM PKE compute takes 134,939 cycles (1.134 ms, 1,951× reduction).

TABLE III
CYCLE COMPARISON VS. 32-BIT RISC-V SOFTWARE [3]

Primitive / Scope	Official HQC-128 SW (32-Bit RISC-V [3])	Our PKE Coprocessor (xc7a35t-1 @119 MHz)	Compute Cycle Reduction
KeyGen	43,592,464 cyc	19,796 cyc (0.166 ms)	2,202×
Encap / Encrypt	87,574,395 cyc	45,677 cyc (0.384 ms)	1,917×
PKE Decrypt (Core Only)	132,101,254 cyc	23,789 cyc (0.200 ms)	5,553×
FO-KEM Decap (Dec + Re-Enc)	132,101,254 cyc	69,466 cyc (0.584 ms)	1,902×
Total (PKE Core Only)	263,268,113 cyc	89,262 cyc (0.750 ms)	2,949×
Total (FO-KEM Core Compute)	263,268,113 cyc	134,939 cyc (1.134 ms)	1,951×

Across the 32-bit AXI4-Lite bus, host transfer of 1,676 64-bit words (845 input words written, 831 output words read) takes 3,352 32-bit register transfers (∼5,028 transactions, or ∼10k–20k bus cycles). In [3], polynomial multiplication consumes 95% of runtime on CV32E40P; host routines (SHAKE256, sampling, control) consume ∼13.2M cycles (≈ 5%). Under Amdahl's Law, hardware acceleration of arithmetic and decoding from ∼250.1M to 134,939 cycles bounds projected SoC speedup to $S_{\text{SoC}} \approx 263.3\text{M}/(13.2\text{M}+0.135\text{M}+0.02\text{M}) \approx 19.7\times$. Compared to ANDROMEDA's measured 9.5× SoC speedup (27.80M cycles, 8.0 BRAMs) [3], our core projects double the speedup with 25.0% fewer BRAMs.

V. CONCLUSION

This letter presented a memory-efficient FPGA co-design of HQC-128 PKE for edge nodes. Modeling polynomial lifetimes as an interval graph ($\chi(G) = 6$) and using distributed LUTRAM fits all three PKE primitives and the RS-RM codec into 6.0 RAMB36E1 tiles (12.0% of an Artix-7 xc7a35t) with 0 DSPs. At 119.0 MHz, the core consumes 0.158 W and 118.5 μJ per sequence, and reduces compute cycles by 2,949× for PKE (1,951× for FO-KEM) with a projected ≈ 20× SoC speedup over 32-bit RISC-V software.

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